

Professional Summary: Computational Materials Scientist with a Master's degree in Physics and a PhD in Theoretical Chemistry & Modelling, with doctoral defense scheduled for **December 11, 2026**. Research experience spanning **electronic-structure calculations, atomistic simulation, and multi-scale modelling** across diverse materials, including **2D materials, porous materials, MOF-polymer composites, perovskites, and spinel oxides**. PhD research focuses on developing **integrated DFT, Monte Carlo, and molecular dynamics workflows** to investigate adsorption, molecular transport, and stability of MOF-polymer composites under application-relevant conditions. Proficient in **DFT, MD, MC, graph-based analysis, statistical modelling, and high-performance scientific computing**, with growing expertise in **AI/ML for materials science** for materials property prediction and data-driven discovery. Interdisciplinary background spanning **quantum chemistry, condensed-matter physics, atomistic and multi-scale simulation, statistical mechanics, and machine learning**.

Research Experience

- **PhD Researcher** - MOF/Polymer Composites for Capture of Volatile Organic Compounds under humidity 2023 - present
CNRS / Université de Montpellier & CNES, France

Developed an integrated multi-scale computational workflow (DFT, GCMC, MD) to design and characterize MOFs/hydrophobic polymer composites for selective VOCs capture under humid conditions relevant to spacecraft environments

- **Visiting Researcher** Aug 2022 - Nov 2022
Universität Duisburg-Essen/CRC 247, Germany

Electronic structure studies of complex materials using quantum methods

- **Master's Researcher** - Electronic structure and properties of 2D materials 2020 - 2021
Theoretical Condensed Matter Physics & Advanced Computational Materials Science Lab, IIT Indore

DFT investigation of Mn-doped MoS₂ monolayers as electrocatalysts for hydrogen evolution; work published in Physical Chemistry Chemical Physics (RSC, 2022)

Publications

- MOF/Hydrophobic Polymer Composites for Toluene Capture. Joy Ekka; Sabine Devautour-Vinot; Guillaume Rioland; Guillaume Maurin (*ACS Applied Material and Interfaces*)
- Framework-Dependent Adsorption Regimes of Linear and Cyclic Siloxanes in Metal-Organic Frameworks under Trace Concentrations. Pedro Freire Brântuas; Joy Ekka; Ezgi Gulcay-Ozcan; Guillaume Rioland; Guillaume Maurin; Sabine Devautour-Vinot (*ACS Inorganic Chemistry*)
- Unveiling the Role of 2D Monolayer Mn-doped MoS₂: Toward an Efficient Electrocatalyst for H₂ Evolution Reaction. Joy Ekka; Shrish Nath Upadhyay; Frerich J. Keil; Srimanta Pakhira (*RSC PCCP*)

Technical Skills

- **Computational & Materials Modelling:** DFT · ab-initio and classical Molecular Dynamics (MD) · Grand Canonical Monte Carlo (GCMC) · Monte Carlo (MC) · Coarse-grained modelling · Statistical mechanics · Graph theory & algorithms · Pore/morphology analysis · Machine-learned interatomic potentials (MLIPs) · Numerical methods · High-performance computing (HPC) · Generative diffusion models
- **Programming & Scientific Computing:** Python · C++ · MATLAB · Linux/HPC environments · Git/GitHub · Workflow automation
- **Simulation & Computational Tools:** · LAMMPS · VASP · CP2K · GROMACS · RASPA · Gaussian · Crystal

Education

- PhD in Theoretical Chemistry & Computational Modelling Dec. 2023 - Dec. 2026
CNRS / Université de Montpellier, France | Co-funded: CNES & CNRS
- MSc in Physics 2019 - 2021
Indian Institute of Technology (IIT) Indore, India
- BSc (Honors) in Physics 2016 - 2019
St. Xavier's College, Ranchi, India

Conferences & Presentations

- MOFSIM 2026: Poster presentation
- COMET-STR 2026: Oral presentation
- E-MRS Spring Meeting 2026: Oral presentation
- EUROMOF 2025: Poster presentation
- CNES Annual Research Meeting JC1 2024 & JC2 2025: Oral & poster presentation, Toulouse
- MOFSIM 2024: Attendee

Languages

English (fluent), French (A2), Hindi (native)